ColdTrust Pitch Day Reader

Setup steps, the word-for-word script, plain-language explanations and the questions judges may ask.

Team Bibilabu · Live pitch on Saturday 7 November 2026: 5 minutes including the video, then 2 minutes of questions · Prepared Wednesday 4 November 2026

What is in this reader

Chapter	What it gives you	Read it
1. Setup steps	Wiring and flashing the board, installing the software, turning on the AI agents, the pitch-day setup timeline, and what to do when something fails	Wednesday and Thursday
2. Final script and evidence guide	The fix list for your deck, the timed script, the Seal agent terms, what each rule decides, every appendix slide word by word, 35 questions with answers, and the five hardest answered in full	Thursday and Friday
3. Video and preparation notes	The video review and new storyboard, the numbers to know, the reading list and the pitch-day checklist	Friday

Review notes: how the two documents fit together

I read the Pitch Day Playbook (written on 1 November) against the Final Script and Evidence Guide (written on 4 November, after your final deck existed). The Guide is the newer one, so where they differ, **follow the Guide**. This reader keeps all of the Guide, and only the parts of the Playbook that are still true.

Finding	What I did
The Playbook's slide plan has seven slides with a "Proof" slide at position 5. Your final deck has "AI Agent Workflow" at 4, the video at 5, "Trust and limits" at 6 and "Thank you" at 7.	Dropped the Playbook's slide plan and script. The Guide's script matches the final deck.
The Playbook's Canva checklist and its "Plan to 18:30" list are dated 1 November and are finished or out of date.	Dropped both.
The Playbook has 16 judge questions. The Guide has 35 and covers all 16.	Kept the Guide's questions only.
Your final deck has no slide for the measured results (0 false alarms against 1, 3 of 3 attacks against 0, cargo held 5 of 5 against 0 of 5, 11 minutes earlier). Appendix A1 (architecture) and A2 (the detection table) from the plan were never added. The numbers appear only when you speak slide 6 and in the video's last card.	Left as the deck stands, and flagged it here. Add an appendix slide for the detection table before Saturday, because "how do you know it works?" is the question you are most likely to get. I can build it in the same style as slides 8 and 9.

Finding	What I did
The Playbook's video review and storyboard are still valid.	Kept as Chapter 3.
The numbers in both documents agree (60 of 60, 0 invented, 0 of 3, 312 tests, 0 vs 1, 3 of 3 vs 0 of 3, 11 minutes).	No change.

Still to do on the deck (the Guide's fix list, repeated because it is the most urgent work): delete the hidden "Vision 02" and Lorem ipsum text on slide 3, make the slide 6 limits specific, stop the slide 6 title clipping at the right edge, equalise the label sizes on slide 4, reword box 3 on slide 2, and make sure the video is 2:00 or shorter.

Chapter 1: Setup steps

What you are setting up

There are three things, and you only need all three if you want to show the hardware live.

Piece	Purpose	Start it with
The sensor node and the camera	The real hardware, as a prop or a live demo	`demo.bat`
The no-hardware demo	A fallback that needs no board, no login, no Telegram and no API key	`judge_demo.bat`
The slide deck with the video	What you actually present	PowerPoint, slideshow mode

Your pitch uses the recorded video, so the hardware is optional on the day. Set it all up anyway, because a judge may ask to see it.

1. Hardware setup (once)

Wiring

Power every module from **3V3**, never 5V: the analog pin must not see more than 3.3 V.

Part	Module pin	NodeMCU pin
DHT11	VCC / GND / DATA	3V3 / GND / **D2**
Light module (SN-LIGHT-MOD)	VCC / GND / AO / DO	3V3 / GND / **A0** / **D5**
Green LED (heartbeat, sent OK)	plus through a 220 to 330 ohm resistor, minus to GND	**D7**
Red LED (alarm, Wi-Fi down)	plus through a 220 to 330 ohm resistor, minus to GND	**D1**
Logitech C270	USB	the **laptop**, not the NodeMCU

Rules: leave D3, D4 and D8 free because they decide how the board boots. Point the LEDs away from the light sensor, or the sensor will read them as "door open". The diagram is `firmware/wiring.svg`, and the tick-box wiring check is in `firmware/README.md`.

Arduino IDE (once)

1. File, Preferences, Additional boards manager URLs: `http://arduino.esp8266.com/stable/package_esp8266com_index.json`
2. Tools, Board, Boards Manager: install **esp8266** by ESP8266 Community.
3. Tools, Board: **NodeMCU 1.0 (ESP-12E Module)**. Pick the COM port that appears when you plug the board in. If no port appears, install the USB driver for the chip printed on the board (CH340 or CP2102), and use a **data** cable, not a charge-only cable.
4. Tools, Manage Libraries: install **DHT sensor library** (Adafruit, accept Adafruit Unified Sensor) and **ArduinoJson** by Benoit Blanchon, version **7.x**.

5. Open the Serial Monitor at **115200** baud.

Settings file

Copy `firmware/secrets.example.h` to `firmware/secrets.h` and fill in:

Setting	What to put
`WIFI_SSID`, `WIFI_PASS`	A **2.4 GHz** network. The ESP8266 cannot join 5 GHz. A phone hotspot works if set to 2.4 GHz or "maximise compatibility".
`SERVER_URL`	`http://<laptop IP>:8000`. Get the IP from `ipconfig`, the IPv4 address on the same Wi-Fi. No spaces inside the quotes.
`DEVICE_SECRET`	The same value as `COLDTRUST_SECRET` in `cold-chain/.env`.

The laptop IP changes when the Wi-Fi changes. After moving to a new network, update `SERVER_URL` and upload the sketch again. `secrets.h` is kept out of GitHub; never commit it.

Test, then flash

Test each part alone first (`firmware/tests/01_dht11` to `06_post`), then open `firmware/firmware.ino` and upload it. The Serial Monitor prints a self-test: `[PASS]` for each part, or `[CHECK]` with the wire to look at. Type `t` and Enter to run it again.

LED	Meaning
Short green blink every 5 s	A sensor reading worked
Double green blink	Data reached the backend
Solid red	Alarm: door open or temperature out of range
Fast red blink	Wi-Fi is down; readings are being kept

2. Software setup (once)

1. Install **Python 3.10 or newer** and tick "Add python.exe to PATH".
2. Get the project: `git clone https://github.com/Serineng510/Cold_Trust.git`, then open the `cold-chain` folder.
3. Create the environment and install: `python -m venv .venv`, then `.venv\Scripts\python -m pip install -r requirements.txt`. The camera agent downloads two small model files the first time it runs.
4. Copy `.env.example` to `.env`. Generate a device key with `python -c "import secrets; print(secrets.token_hex(32))"` and put it in `COLDTRUST_SECRET`. The backend refuses to start on real hardware with the default key.
5. Optional: add the Telegram bot token and chat IDs. `python -m backend.telegram_setup chats` lists the chat IDs.
6. Allow Python through Windows Firewall on **private networks** when it asks the first time; the board must reach port 8000.

3. Turning on the AI agents (optional)

Without a key everything works and the screens show the rules' own text.

1. Create a free key at aistudio.google.com/apikey.
2. In `.env`, add `GEMINI_API_KEY=your-key`. Leave `COLDTRUST_VISION` off so camera frames never go to Gemini.
3. Check it: `.venv\Scripts\python -m backend.gemini_check`. You want `[ok]` for each agent.
4. Restart the server.

The free tier allows 15 requests a minute and 500 a day, and the backend keeps itself under 60% and 80% of those. If a call fails, the rules' text is shown. Never put your key in a slide, a screenshot or GitHub.

4. Running the demos

Want	Double-click	Notes
Hardware demo	`demo.bat`	Starts the server and the camera in two windows. Wait for **Live** at http://127.0.0.1:8000. Add `-NoCamera` if no camera is plugged in.
Real readings only	`official.bat`	Mock chips are refused. The real DHT11 may raise freeze alerts.
Stop everything	`stop.bat`	
What is running?	`status.bat`	Two servers at once cause blank or flickering video.
No-hardware fallback	`judge_demo.bat`	Add a scenario name: `spoof` (default), `theft`, `door_ajar`, `compressor_fail`, `authorised_stop`, `freeze`, `stopped_theft`, `replay`, `sensor_disagree`, `normal`.

Theft story with the hardware: in the dashboard, tick Mock on the Light chip, stand in front of the camera, and press Activate Route with a speed above 0 so the truck is away from a stop. Then show the Evidence tab. Untick the mocks when you finish.

Port 8000 busy? Close the other program, or start the fallback on another port: `judge_demo.bat spoof -Port 8001`.

Never reset or demo with `data/official.db`. The scripts already keep demo and official data apart.

5. Pitch-day timeline

When	Do this
Wednesday 4 Nov	Fix the deck (Chapter 2 fix list). Re-cut the video to 2:00 or less with captions. Run `judge_demo.bat` once with internet so the first-run setup is done.
Thursday 5 Nov	Export the deck as PPTX **and** PDF. Check the video plays inside the PPTX. Add the detection-table appendix slide. Rehearse the script aloud three times with a timer.
Friday 6 Nov	Open the PPTX on a different computer and play slide 5 with sound. Copy the PPTX, the PDF and the MP4 to a USB stick and to cloud storage. Charge everything. Pack a display adapter, a clicker, the board, the USB cable and a phone hotspot. Rehearse twice more. Print the numbers card.
Saturday 7 Nov, 30 minutes before	Plug in the laptop. Turn off notifications (Focus assist) and close Telegram. Set sleep to never. Connect the display and choose Duplicate. Open the deck in slideshow mode, play the video once to check the sound, then return to slide 1.

When	Do this
If showing the board	Power the board, run `demo.bat`, wait for **Live**. Keep it minimised.
Always	Start `judge_demo.bat spoof -Port 8001` in a minimised window as a live fallback.
During questions	Type a slide number and press Enter to jump to an appendix slide (8, 9 or 10).
After	Run `stop.bat`.

6. If something fails

Problem	What to do
The video does not play in the deck	Play the MP4 from the desktop in the Movies and TV app, then come back to the slide.
The deck will not open	Open the PDF copy.
The projector shows nothing	Press Windows key and P, then choose Duplicate.
"Access denied" on the COM port	Close the Serial Monitor and anything else using the port, unplug and replug the board, try another data cable.
The board cannot reach the dashboard	Check you are on the same 2.4 GHz network, update `SERVER_URL` to the laptop's current IPv4 address, upload again, and check the firewall.
Port 8000 is busy	Run `stop.bat`, or choose another port with `-Port 8001`.
The AI is silent or slow	Say "the rules explain when the AI is unavailable", which is the design. Do not apologise.
Map tiles are blank	They need internet. The decision logic does not.
You run out of time	Skip the second half of slide 6's limits. Never skip slide 7.

Chapter 2: Final script and evidence guide

Your deck is close: the story and order are right, but the script needs to fit 5 minutes with the video, two slides still carry template leftovers or vague text, and three terms in the Seal agent need a plain-language answer you can say out loud.

Fix before 7 November

I rendered all 10 slides of `ColdTrust_TeamBibilabu.pptx`. Six things to fix, most important first:

☐ **Slide 3 has hidden template text.** "Vision 02" and a Lorem ipsum paragraph sit faintly behind the Human Oversight box and near the top. Click the faint area, press Delete, and check with Selection Pane.

☐ **Slide 6 limits are still vague.** "DHT11 measurement limitations" and "Simulated testing" invite a hard question. Use: "Simulated tests by our own team; small sample"; "DHT11 reads whole degrees; light approximates a door; no second thermometer; GPS simulated"; "Free-tier AI; records not GS1-certified". Change " photo evidence" to "Photos hashed (SHA-256); records structured to GS1 EPCIS 2.0".

☐ **Slide 6 title touches the right edge.** Shrink it a little so "limits" is not clipped.

☐ **Slide 4 labels differ in size.** "Verified data" is large and "Administrator approves" is small; make all four the same size. Remove the dashed line from step 3 to the Thermal and Incident agents.

☐ **Slide 2 box 3 wording.** "Identifies potential security risks by CISA" should read "CISA (2020) flags vehicle hijacking and cargo tampering; monitoring data must be protected against manipulation".

☐ **Slide 5, the video.** The script below assumes 2:00. If your video is longer, cut it, or every slide after it loses time.

The appendix is right: slide 8 is the AI evaluation, slide 9 the Seal checks and approval menu, slide 10 the hardware setup. Part 3 explains all three.

The 5-minute script

It finishes at 4:43, which leaves 17 seconds of slack. That only holds if your video runs 2:00 or less. Speak at a steady pace of about 150 words a minute, and pause after each slide's last sentence before you advance.

Slide	Content	Start	Length
1	Title	0:00	0:12
2	Problem statement	0:12	0:28
3	The solution	0:40	0:35
4	AI agent workflow	1:15	0:33
5	Demo video	1:48	2:00
6	Trust and limits	3:48	0:40
7	Thank you	4:28	0:15

Slide 1, title (12 s)

> "Good afternoon. We are Team Bibilabu, and this is ColdTrust. A truck of vaccines can arrive spoiled while its sensor log says everything was fine. ColdTrust checks its own data before it acts."

Slide 2, problem statement (28 s)

Point at each box as you name it.

> "Cold chains have three problems. Research found temperature swings, reliance on data loggers and manual checks, and a skills gap with pharmaceuticals. The result is unnecessary alarms, slow reaction to cooling failures, and alerts nobody can explain. And CISA warns of hijacking and cargo tampering, so the monitoring data itself must be protected. Our users are the administrator, who decides, and the driver, who sees alerts read-only."

Slide 3, the solution (35 s)

Read the boxes left to right, then bottom.

> "ColdTrust joins hardware and software. A sensor node and camera send signed readings to a verification gateway. It checks the HMAC signature, sequence number, freshness and replays, and logs every check in a Seal trust report. Then rules set each package's status and risk score. Three Gemini agents explain and recommend, but cannot change a status. Administrators approve quarantine, diversion or taking the vehicle offline. Theft triggers quarantine for their review, and if the AI fails, the rules explain."

Slide 4, AI agent workflow (33 s)

Walk along the four steps with your hand, then point at each agent.

> "The workflow has four steps. Data is verified. Rules set the status. Then three agents explain and propose. The Thermal agent explains temperature deviations and flags a possible cooling failure, with a maintenance report only after at least sixty readings. The Seal agent explains data-integrity problems and which checks failed. The Incident agent weighs thermal and security risk, proposes an action from a fixed menu, and words the draft report. Finally, the administrator approves."

Slide 5, demo video (2:00)

Say one line, press play, then stay silent and look at the judges, not the screen.

> "Here is ColdTrust running."

Slide 6, trust and limits (40 s)

> "Our safeguards: signed sensor data, replay and freshness checks, AI that cannot change a status, admin-only approvals, hashed photo evidence, and people detected but never identified. Our limits:

> simulated tests by our own team, a DHT11 that reads whole degrees, and Gemini's free tier. Even so, across ten simulated scenarios, ColdTrust raised no false alarm where a plain alarm raised one, recognised three of three data attacks where the alarm saw none, and alerted eleven minutes earlier on a failing cooler. Next is a field trial on a real truck."

Slide 7, thank you (15 s)

> "ColdTrust does not just alarm. It checks that its own data can be trusted, then helps a person act. You can try it without hardware: run judge_demo.bat from our GitHub. Thank you, we welcome your questions."

During the 2 minutes of questions

Keep slide 7 up. To open an appendix slide, type its number and press Enter during the show:

Judge asks about	Type	Slide
AI hallucination, injection, how the AI was tested	8	AI evaluation
Faked data, signatures, who approves actions	9	Seal checks and approval menu
What hardware you actually used	10	Hardware setup

Part 1: the Seal agent's terms in plain language

The idea in one minute. The sensor node packs its readings into a message, adds a secret-key fingerprint, numbers the message and stamps the time. The backend rebuilds the fingerprint, checks the number is the next one, checks the readings are newer than anything stored, and checks the clock. Only then are the readings used. The Seal trust report is the list of those checks.

Hash

A hash is a fingerprint of data made by a one-way function. SHA-256 turns any message into 64 hexadecimal characters. Change one character of the message and the fingerprint changes completely, and you cannot work backwards from the fingerprint to the message. A hash alone proves nothing about who made it, because anyone can compute one.

HMAC and HMAC-SHA256

HMAC means Hash-based Message Authentication Code: a hash mixed with a **secret key**. HMAC-SHA256 is HMAC built on SHA-256. The node and the backend both hold the same secret key. The node computes the code over the message, and the backend recomputes it. If the two match, the message came from someone holding the key and nobody changed it on the way. An attacker who does not have the key cannot produce a matching code, even knowing the algorithm.

Signature

The signature is the HMAC code sent along with the message. The Seal card line "HMAC-SHA256 matches the device key (signature d0742cb6...)" means: the backend recomputed the code and got the same value; `d0742cb6` is just the first 8 of its 64 characters. If a judge says "that is not a real digital signature", agree: it is a shared-key

message authentication code, not a public-key signature. It is the right tool here because one device talks to one backend, and it is far cheaper for an ESP8266.

Signed sensor data

The readings travel inside the signed text, together with the identifiers, in this order: device ID, sequence number, timestamp, message type, trigger, then each reading as time, temperature, humidity and light (plus a second temperature if fitted). Because all of it is signed, changing a temperature, a time or the sequence number in transit breaks the signature and the message is rejected and logged.

Sequence number

Every message carries a counter. It starts at the node's boot time, so it keeps rising even after a reboot. The backend remembers the last one and compares:

New number against the last	Result	Meaning
Not higher	Replay, message rejected	Someone resent an old message
Exactly one higher	Passed	Normal
More than one higher	Suspicious (a gap)	Messages were lost or dropped

The dashboard shows it as "1793512525162 follows 1793512525161".

Freshness

Freshness is two separate checks on the card:

- **New readings.** The newest reading must be later than everything already stored. If not, it is a replayed batch.
- **Time window.** The message timestamp must be within 5 minutes (300 s) of the server clock. The node gets real UTC time from the internet and logs nothing until its clock is set. This shows "not run" in the simulator, because simulated time is not real time.

Why you need all three

A replay attack records a genuine, correctly signed message, such as "temperature normal", and resends it later while the door is open. Its signature is still valid, so the signature alone cannot catch it. The **signature** stops forgery and editing, the **sequence number** stops resending, and **freshness** stops old data being passed off as new.

What this does not cover, and what does

A valid signature proves the message is authentic, not that the sensor is telling the truth. If someone holds ice on the DHT11 or freezes a reading, the message is still genuine. That is why the Seal agent has more checks: light jitter, door against temperature, humidity response and a second thermometer. In your spoof demo the signature, sequence and new-readings checks all pass, yet the result is **Not trusted** because the door was open for over 5 minutes in warm air while the temperature never moved. Say this out loud: "authentic does not mean true". The secret key itself lives in `firmware/secrets.h`, which is kept out of GitHub, and the server refuses to start on real hardware with the default key.

Reading the card

Each check shows passed, suspicious, failed or not run. The overall result is **Not trusted** if any check failed, **Suspicious** if any check is suspicious, and **Trusted** otherwise; a check that did not run does not count against the

result. In your spoof screenshot, 6 of 8 checks ran: signature, sequence, new readings and light jitter passed; door against temperature failed; humidity response was suspicious; time window and second thermometer did not run. So the result was Not trusted.

One sentence for each, if a judge asks

Term	Say this
Hash	A fixed-size fingerprint of the data; change one character and it changes completely
HMAC-SHA256	A hash mixed with a shared secret key, so only the real node can make a matching code
Signature	The code attached to each message; the backend recomputes it and compares
Signed sensor data	Readings, time and counter all sit inside the signed text, so none can be edited
Sequence number	A counter that must go up by one; a repeat is a replay, a jump is a gap
Freshness	The readings must be newer than stored ones and within 5 minutes of the server clock

Part 2: what the rules decide

"Rules decide" means ordinary code, with fixed thresholds, makes every decision that changes cargo. The AI never does. The rules run in five stages, and each stage decides one thing.

Stage 1: is this message accepted?

Rule	It decides	Threshold
Signature	Accept or reject the message	HMAC-SHA256 must match the device key
Sequence number	Accept or reject	Must be higher than the last one
Timestamp	Accept or reject (real hardware only)	Within 5 minutes of the server clock
Readings newer than stored	Accept or reject	Newest reading later than everything stored

A rejected message is logged and nothing from it is stored, so it can never change a status.

Stage 2: what is wrong with the readings?

These are the findings the rules raise from accepted readings.

Rule	It decides	Threshold
Door open	Whether the hold is lit	Light at 30% or more
Out of limits	Whether a temperature problem exists	Outside any package's limits for 2 readings in a row
Rate of change	Whether it is warming or cooling too fast	0.5 C per minute or more
Step change	Whether there was a sudden jump	5 C within about 60 seconds
Flat light	Whether the light sensor is frozen	No variation across the last 20 readings
Door with no temperature response	Whether the data is contradictory	Door open over 300 s in 31 C air, temperature unchanged

Rule	It decides	Threshold
Humidity response	Whether humidity agrees with the door	Humidity moved 3% or more while temperature stayed flat
Second thermometer	Whether two sensors agree	More than 3 C apart, 3 readings in a row

Stage 3: what situation is this?

The rules combine the findings with context: speed (above 8 km/h counts as moving), the zone, whether the route is active, and the camera.

Situation	Verdict	Effect on cargo
Light, plus a person seen by the camera, outside an authorised stop	Theft suspected	Every package quarantined, custody tokens revoked
Light while stopped at 0 km/h on an active route, outside an authorised stop	Theft suspected	Same
Light, but the temperature does not react for 5 minutes in hot weather	Spoof suspected	Under review
Light at an authorised depot or dock	Authorised handling	Logged only; becomes an alert after a 10-minute grace if the hold stays out of limits
Light anywhere else, such as driving with the door open	Door ajar	Under review
Temperature problem, rising or falling	Thermal excursion or freeze risk	Decided per package
Flat or disagreeing sensor data	Data trust	Alert only, no status change, because one weak signal must not block cargo

Two details that show the rules are careful. An open door at an authorised stop never excuses a cooling hold, because opening a door does not cool anything. And a camera sighting only counts for 120 seconds.

Stage 4: how risky is each package?

Every package has its own temperature limits and an excursion budget, the minutes it may spend outside its limits. The risk score runs from 0 to 100:

Part	Points
Share of the package's budget used (capped at 100%)	up to 50
Peak deviation, scaled so 10 C counts as the maximum	up to 20
Light exposure on a light-sensitive product	15
Rate of change, scaled so 2 C per minute is the maximum	up to 10
Data trust is low	5

Tamper verdicts also set a minimum score: theft 90, spoof 50, suspicious person 40, door ajar 35. A stolen-from hold is not low risk just because it is still cold.

Stage 5: what is the package's status?

Status	When the rules set it
Quarantined, do not accept	Theft suspected, or the whole budget is used, or risk is 70 or more
Under review	Risk is 30 or more, or the package is out of range now, or the verdict is door ajar, spoof, thermal excursion or freeze risk
Normal	None of the above

Automatic changes only move a status up. A package under review returns to normal by itself only when risk falls below 30 and the hold is no longer lit. A quarantine is lifted only by a supervisor override with a written reason, which also issues a new custody token.

What a custody token is

Each package has a token. A receiving scanner calls the backend with it and is told whether the package is valid to accept. Quarantine revokes the token, so the scanner says "not valid" with the reason, and the old token stays dead even after release.

Rules around the edges

- **Camera:** at an authorised depot or dock a person is logged as handling; anywhere else it is a suspicious person (alert, cargo under review); anywhere else with the hold lit it is theft.
- **Notifications:** a message goes out only when a problem is new, escalates, changes a package's status, or is still unresolved after 10 minutes. An all-clear resets this.
- **Watchdog:** alerts if the node is silent for 30 minutes or the GPS is stale for 10 minutes.
- **Approvals:** the rules decide which of the four actions are offered; only an administrator approves; an unanswered proposal expires after one hour.
- **Reports:** theft and suspicious-person reports are drafted automatically about 15 seconds after the alert. A temperature event gets a report only after an AI maintenance review shows the cooling unit may be weakening, from at least 60 readings, covering the time of the event.

What the rules never do

The AI cannot set a status, release cargo or run an action. It writes explanations, proposals and report wording, and every number in its text is checked against the data.

Part 3: the appendix, word by word

Your deck has three appendix slides. Slide 8 proves the AI behaves, slide 9 shows how data is trusted and who approves, slide 10 shows the real hardware. Each one below says what every part means, where the number comes from, and what a judge may ask.

Slide 8: "60 live AI calls, checked before any safeguard touched them"

What the title means. You made 60 real calls to Gemini. "Checked before any safeguard touched them" means the figures are the model's raw answers, so they show how the model behaves, not what your filters hid.

The setup line. `gemini-3.5-flash-lite free tier` is the model. `10 scenarios x 3 runs` means ten simulated situations from the simulator, each run three times, which gives 30 alert analyses. Each alert also got one report-

wording call, which gives 30 more. 30 plus 30 is 60. `21 October 2026` is the date of the run.

Row	What it means	Where the number comes from
Usable answers: 60 of 60	No error, no quota block, no unreadable reply. If one fails, the app shows the rules' own text instead.	`docs/agent_eval.md`
Invented numbers: 0 of 60	Every figure the AI wrote also appeared in the data it was given. A number check drops the whole answer otherwise.	Same
Actions outside the fixed menu: 0 of 9 Incident answers	Only the Incident agent proposes actions. Three incident scenarios times three runs is 9. None proposed anything outside the four-action menu.	Same
Hidden instruction obeyed: 0 of 3	A prompt-injection test: an instruction was planted inside a finding and a product description, three runs. None of the agents followed it.	Same
Report wording claiming damage or certain theft: 0 of 30	In 30 report-wording calls, the AI never stated cargo damage or said theft was certain.	Same
Response time: median 1.6 s, slowest 7.1 s	Time per alert-analysis call. Median means half were faster.	Same

The card, "Found and fixed by this evaluation". (1) When report wording and action proposals were in one call, 4 of 9 Incident replies ran away into thousands of characters and gave no answer, so the wording became a separate call. (2) One answer contained a stray control character (a NUL byte), which is illegal in a Word file and could corrupt the report, so every AI text is now cleaned. Both have automated tests.

The footer. "Evidence, not a statistical guarantee": 30 and 30 answers on simulated cases are small, and the checks are automatic and narrow. They do not score how helpful an explanation is.

Questions it answers, and your reply.

- *Does the AI make things up?* No invented number in 60 calls, and the number check removes any that would appear.
- *Is 3 injection tests enough?* No, it is a demonstration. The real protection is the design: AI text never sets a status or releases cargo, and actions come from a fixed menu.
- *Why free tier?* It is enough for a prototype; a private endpoint is the next step.

Slide 9: "How data is trusted, and what an administrator can approve"

Left table, the eight Seal checks. Each shows passed, suspicious, failed or not run.

Check	What it asks, in plain words	Why it exists
Signature	Does the HMAC-SHA256 code match the device key?	Stops forged or edited messages
Sequence number	Is this the next message in the count? A gap is suspicious, a repeat is rejected.	Stops replays and shows lost messages
New readings	Are the readings newer than everything stored?	Stops an old batch passing as new
Time window	Is the message time within 5 minutes of the server clock?	Stops delayed or stale data. Not run in the simulator

Check	What it asks, in plain words	Why it exists
Light jitter	Does the light reading vary across the last 20 readings?	A real sensor is never perfectly constant, so a flat line means frozen or replayed
Door against temperature	Was the door open over 5 minutes in warm air while the temperature never moved? If so, it fails.	Physically impossible, so the data is probably fake
Second thermometer	Do two thermometers agree within 3 C?	Catches one sensor being faked or failing. Not fitted on your prototype, so it reads not run
Humidity response	When the door opens, did humidity change? Humidity moving while temperature stays flat is suspicious; neither moving is failed.	A second witness for the door. The 3% threshold is not yet calibrated on the hardware, and the card says so

Under the table. "Result: trusted, suspicious or not trusted." Any failed check gives Not trusted, any suspicious check gives Suspicious, otherwise Trusted. "A check that cannot run says not run; it is never skipped silently" means a missing check is shown, not hidden, so the result is never overstated.

Right card, the approval menu. The Incident agent may only propose these four. Each is offered only when the rules allow it:

Action	What it does	When the rules offer it
Quarantine the cargo	Marks packages quarantined and revokes their custody tokens	Any alert type, while some package is not yet quarantined
Divert to the backup cold store	Sends the cargo to Seremban Cold Store	Thermal problems only (excursion or freeze risk)
Take the vehicle offline	Takes the vehicle out of service	Thermal or security alerts, if it is not already offline
Prepare the incident report	Drafts the Word report	Security events always; a temperature event only after a maintenance review supports it

The note. "Only an administrator approves; the driver is read-only." A proposal is capped at three actions, its reason is capped at 200 characters and shown as text only, never run. An unanswered proposal expires after one hour, and the same one is not raised again for 10 minutes after a decision.

Questions it answers, and your reply.

- *Can someone just fake the sensor data?* The signature stops editing in transit. If they fake the sensor itself, the contradiction checks catch the impossible behaviour; in the demo that gave Not trusted.
- *Who controls disruptive actions?* Only the administrator, from a fixed menu the rules narrow.

Slide 10: "Hardware / Overview / Setup"

Left photo, "Hardware", is the sensor node. Right photo, "Overview", is the whole prototype on your desk.

Part	What it is and what it does
ESP8266 NodeMCU	The small Wi-Fi board that reads the sensors, signs the messages and sends them. It keeps one hour of readings in memory when Wi-Fi is down.
DHT11 (white block)	Reads temperature and humidity. Accurate to about 2 C and whole degrees only, so it is coarse.

Part	What it is and what it does
Light module (blue board)	Reads light level. A lit cargo hold means the door is open. It approximates a door sensor.
Green LED	Heartbeat: a reading was read or delivered
Red LED	Alarm, or Wi-Fi is down
Logitech C270 webcam	On top of the laptop. MediaPipe finds people in the picture and never says who.
Laptop	Runs the backend, the rules, the agents and the dashboard

How it behaves: the light is checked every 0.1 s, temperature and humidity every 5 s, a reading is saved every 30 s, and an event such as the door opening is sent immediately. Be honest that this is a breadboard prototype wired to a laptop, not yet in a truck, and that GPS and the second thermometer are simulated or missing.

Question it answers: *Is this real hardware?* Yes: the ice-on-sensor clip in your video is the real DHT11, and the dashboard in the right photo shows live readings from it.

If a judge asks for the evidence pack you uploaded

Pack page	What it is	One line to say
2	Problem, users, solution	Taken from the submitted proposal
3 and 4	Architecture and agent workflow	The same flow as slides 3 and 4
5	Hardware photos	The same photos as slide 10
6	Measured result against a plain alarm	0 false alarms against 1, 3 of 3 attacks against 0, 11 minutes earlier; simulated
7	AI evaluation	The same figures as slide 8
8 and 9	Dashboard screenshots	Spoof caught, and theft with its automatic report
10	Sample incident report	An automatic draft, not yet reviewed by a person; photos are labelled samples
11	GS1 EPCIS 2.0 record	Seven event records, hashes recomputed and matched; demo company prefix, not certified
12	Tests, safeguards, limits	312 automated tests passing; limits stated plainly

Part 4: questions judges may challenge

Three judges and two minutes means about four questions, so expect the hard ones, not the easy ones. Answer in one sentence, give one proof, then stop. When the point is a real weakness, agree first, then say what you would do. The "Slide" column says what to type and press Enter to show your evidence; "none" means answer from memory.

The five you must answer perfectly

1. Your results are simulated and written by you, so why should we believe them?

2. A valid signature only proves the message is genuine, so what if the sensor itself lies?
3. What does the AI actually do if the rules decide everything?
4. Is this really agentic, or just a chatbot on top of alarms?
5. Who would use this, and what stops a false quarantine?

The five answers in full

Each answer has the sentence to say aloud (about 30 seconds), the proof to point at, what a judge may push on next, and what to avoid. Say the short version first and only go deeper if they ask.

Question 1. "Your results are simulated and written by you, so why should we believe them?"

> "You are right to ask. The comparison is simulated: ten scenarios we wrote, run against a deliberately fast threshold alarm. It shows what our rules do on known cases, not how often real incidents happen. That is why everything is reproducible with one click, and why a field trial on a real truck is our next step."

Proof to point at

- Both systems receive exactly the same simulated messages, so the comparison is like for like.
- The baseline is generous on purpose: an instant alarm with no delay is the fastest a threshold alarm can be.
- The results are not all wins. On a door event both systems know at the same moment. What differs is what happens next: ColdTrust holds the cargo and names the situation, the alarm only rings.
- The detection comparison uses no AI calls, so it is deterministic and repeatable. Run `judge_demo.bat` and the same cases play out.
- Not everything is simulated: the ice-on-sensor clip uses the real DHT11, and the 60 AI calls were real calls to Gemini, although their inputs were simulated cases.

If they push

- "*Could you have fitted the rules to the scenarios?*" Say yes, that is the limitation: the scenarios are known cases the rules were designed for. A field trial would test cases you did not anticipate. The thresholds are settings that anyone can change and re-run.
- "*Why no real data?*" There was no refrigerated truck during the hackathon. The node, the camera and the dashboard are real, and the next step is a trial.

Avoid saying "proven", "validated" or "tested in real deployments". Agree first, then give the plan.

Evidence: `docs/detection_results.md` (every scenario and time) and `judge_demo.bat`. Your deck has no slide for this yet; add the detection-table appendix slide so you can jump to it.

Question 2. "A valid signature only proves the message is genuine, so what if the sensor itself lies?"

> "Correct: a signature proves who sent a message and that nobody changed it, not that it is true. So the Seal report goes on to ask whether the readings are physically believable. In our demo the door was open for over five minutes in warm air and the temperature never moved. The signature passed, but the data was marked Not trusted."

How the defence is layered

1. **Authentic and new:** signature, sequence number, freshness and time window. These stop forgery, editing, replay and stale data.
2. **Believable:** light jitter (a real light sensor is never perfectly flat), door against temperature, humidity response, and a second-thermometer comparison.
3. **In context:** the rules treat weak data signals carefully. A data-trust problem on its own raises an alert but does not quarantine cargo, because one weak signal must not block a shipment. Combined with a door event it becomes "Spoof suspected" and puts packages under review.
4. **A person:** the Seal agent's advice in the demo log is "Inspect the sensor node physically. Do not rely on this device's data." An administrator decides what happens next.

Be honest about the gaps

- An attacker who feeds a believable temperature and keeps the light consistent with it would pass these checks.
- The prototype has one DHT11, no tamper switch, and **no second thermometer** (that check reads "not run"). The humidity threshold is not yet calibrated on the hardware.
- A stolen device key would let someone forge genuine-looking messages.

Next steps to name: a second, independent thermometer; a reed-switch door sensor; a tamper switch; per-device keys with rotation; and a secure-element chip.

If they push: "*So it can be fooled?*" Say: "Any system can be fooled by a careful enough attacker. Our aim is to make cheap attacks, such as freezing a reading, replaying a message or ice on the sensor, produce a visible Not trusted result, and to show what failed."

Avoid "unhackable" and "cannot be fooled". **Evidence:** the spoof clip in the video, and appendix slide 9.

Question 3. "What does the AI actually do if the rules decide everything?"

> "Rules decide what happens to the cargo, so the system protects it even with no AI. The AI does what rules cannot: it explains the situation in plain language, proposes the next action from a fixed menu, and words the incident report. That matters because the research found a skills gap: people struggle to understand why an alert fired."

What each agent does, with an example

- **Thermal agent:** explains a temperature deviation and, from the sensor history, writes the maintenance review that says whether the cooling unit may be weakening. That review is what allows a temperature incident report to be offered at all.
- **Seal agent:** turns failed checks into a sentence a person can act on, such as "the sensor data contradicts itself, so inspect the node physically".
- **Incident agent:** weighs thermal and security risk, proposes up to three actions from the four-action menu (for theft it typically proposes taking the vehicle offline and preparing the report), and words the draft report.

What limits the AI

- It cannot set a status, release cargo or run an action.
- Every number in its text is checked against the data. In 60 live calls there were 0 invented numbers.
- Its proposals pass through a filter that keeps only allowed actions, drops repeats and caps the reason at 200 characters, and the reason is shown as text, never run.
- If the AI is down or its answer is invalid, the rules' own explanation appears instead.

One nuance worth saying. The AI's maintenance review decides whether a report for a temperature event is *offered*. It never decides whether cargo is held, and a person still approves the report.

If they push: "*What does the AI add that you could measure?*" Be honest: the checks measure safety, not usefulness. Explanation quality and the time saved when reviewing an incident have not been measured yet.

Avoid saying the AI "decides" or "controls" anything. **Evidence:** appendix slide 8 (60 of 60 usable, 0 invented numbers, 0 of 9 actions outside the menu, 0 of 3 injections obeyed) and slide 4.

Question 4. "Is this really agentic, or just a chatbot on top of alarms?"

> "A chatbot waits for someone to ask. Our agents start by themselves when an alert fires. Each one reads the evidence, reasons about it and proposes concrete actions from a defined set of tools. We limited their autonomy on purpose, because a wrong disruptive action on vaccines or food is costly, so a person approves anything disruptive."

What makes it agentic

- **Triggered, not prompted.** One agent call runs per alert, with no one typing a question.
- **Real inputs.** Each agent receives the Seal report, the speed, zone and route, the camera sighting and the product limits.
- **Structured outputs.** The answer has a fixed shape: an explanation, an impact, and proposals that each name an action and a reason.
- **A multi-step workflow.** Verify the data, rules set the status, agents explain and propose, an approvals queue waits, an approved action is carried out by code, and the result is recorded (audit log, EPCIS record, Word report). A theft report drafts itself about 15 seconds after the alert, and the maintenance review feeds a report gate.
- **Autonomy matched to risk.** The system acts on its own where it is safe (alerting, explaining, drafting a report) and asks where it is costly (diverting, taking a vehicle offline).

Be honest about the limits. Each agent makes one decision pass per alert. It does not plan over many steps, call tools freely or loop to improve its own answer. Call this "bounded agency with a human in the loop", and say that you chose it.

If they push: "*Wouldn't more autonomy be better?*" Say: "Not yet. Our evaluation shows the agents behave, but 60 calls on simulated cases is not enough to let them act on real cargo. More autonomy would follow field evidence."

Avoid defending the word "agentic" for its own sake. **Evidence:** slide 4, the approvals clip in the video, and appendix slide 9 (the approval menu).

Question 5. "Who would use this, and what stops a false quarantine?"

Who uses it. The **administrator** (an operations or quality manager at a carrier or a distributor of vaccines, insulin, fish or dairy) monitors shipments, approves actions and generates reports. The **driver** receives alerts in a read-only view. Who pays, and what one node costs, are not in your project files, so prepare your own two sentences. Do not invent a price.

> "The user is the operations or quality manager who has to decide when cargo may no longer be accepted. A quarantine here is a hold, not a loss, and several things stand between a normal event and a quarantine."

What stands between a normal event and a quarantine

1. **Context first.** A door opening at an authorised depot or dock, below 8 km/h, is logged as handling. It becomes an alert only after 10 minutes outside limits, and it never excuses a cooling hold. In the simulation this gave 0 false alarms where the plain alarm gave 1.
2. **More than one signal for theft.** Theft needs light plus a person on camera outside an authorised stop, or light while stopped on an active route outside an authorised stop. A weak data-trust signal alone never quarantines cargo.
3. **A hold, not a discard.** Quarantine marks the packages "do not accept" and revokes their custody tokens, so a receiving scanner is told the package is not valid.
4. **A person reviews.** The administrator reviews the alert. Only a supervisor override with a written reason lifts a quarantine and issues a new token.
5. **A record.** Every status change, notification and approval is in the audit log.

Be honest about the gap. In the field the false-quarantine rate is unknown. A driver who legitimately opens the door outside a depot, for example to check the load, would put the cargo under review, and the driver is read-only and cannot clear it. A reason-and-acknowledge step for the driver would be a sensible next improvement.

If they push: "*Isn't automatic quarantine too aggressive?*" Say: "It is deliberately conservative. A wrongly held shipment is cheap to release; a stolen or spoiled one is not. And we keep the override with a supervisor."

Avoid quoting a price or a market size you cannot source. **Evidence:** slide 3 (Human Oversight), appendix slide 9, and the rules in Part 2 of this reader.

1. Value and problem

Question	Answer	Slide
Is this just a threshold alarm with a chatbot?	No. It verifies the data first, rules then set each product's status from its own limits, and AI only explains. Against a plain alarm in ten simulated scenarios: 0 false alarms against 1, 3 of 3 attacks recognised against 0, and 11 minutes earlier on a failing cooler.	none
Why use AI if the rules decide?	Rules are predictable but cannot explain themselves to a non-expert. The agents put the findings in plain words, propose actions and word the report, which answers the skills gap in the research. If the AI is down, the rules' own text takes over.	8
How do you know the problem is real?	Abu Hassan, Ngah and Ab Talib (2021) found temperature swings, reliance on data loggers and manual checks, and insufficient expertise. CISA (2020) warns about hijacking, cargo tampering and manipulated monitoring data. Say plainly that the paper has no theft content, so the security argument rests on CISA.	none
How is it different from existing monitoring?	Loggers and IoT dashboards record and alarm. ColdTrust also checks that the data can be trusted and explains it with checked AI. You have not benchmarked specific commercial products, so do not claim you beat them.	none
Who uses it, and who pays?	Administrators at carriers or distributors decide; drivers see alerts. Prepare your own answer on the buyer and the node's cost. These are not in your project files, so do not guess a price.	none

2. Data trust and security

Question	Answer	Slide
Isn't an HMAC just a password? Is it a real signature?	It is a shared-key message authentication code, not a public-key digital signature. It proves the message came from a holder of the key and was not edited. One device talking to one backend does not need public-key cryptography, and it is cheap enough for an ESP8266.	9
What if someone records a valid message and replays it?	The signature would still be valid, so two more checks catch it. The sequence number must go up, and the readings must be newer than everything stored, within 5 minutes of the server clock. A repeat is rejected.	9
What if the sensor itself is faked, such as ice on the DHT11?	Agreed, the message is then genuine. That is why the Seal report has contradiction checks. In the demo the door was open over 5 minutes in warm air while the temperature never moved, so the result was Not trusted even though the signature passed.	9
What if the device key is stolen?	An attacker with the key could forge messages. The key sits in a file kept out of GitHub, and the server refuses the default key on real hardware. Next: a per-device key, rotation, and a secure element chip. Physical theft of the node is outside this prototype.	none
Is the time window a problem if the clock is wrong?	The node takes real UTC time from the internet and logs nothing until its clock is set. The window is 5 minutes. It shows not run in the simulator because simulated time is not real.	9
What if messages are lost and the sequence has a gap?	A gap is marked suspicious, not rejected. The node keeps one hour of readings and resends them, oldest first.	9
Is the Seal agent doing the checking?	No. The checks are rules in ordinary code. The Seal agent only explains the result in plain language and shows which checks failed.	4
Where do pictures and data go?	People are detected on the laptop and are never identified. Camera frames are not sent to Gemini unless you opt in, and that is off by default. Alert photos do go to the Telegram group, so use a private group. The free Gemini tier may use submitted text, so only demo data is sent.	none
How are logins protected?	An administrator and a driver sign in with a password and are locked out after repeated failures; the driver is read-only. The judge demo turns sign-in off on purpose.	none

3. Rules and logic

Question	Answer	Slide
How did you choose the thresholds, such as 30% light, 5 minutes, 0.5 C per minute?	They are starting points set from the product limits and testing on the prototype, all adjustable in settings. They have not been calibrated in a field trial, and the humidity threshold says so on the card.	9
Is the risk score arbitrary?	It is a transparent weighted sum: up to 50 points for budget used, 20 for peak deviation, 15 for light exposure, 10 for rate, 5 for low data trust. The weights are a stated starting point and have not been sensitivity-tested.	none
How do you avoid false alarms at a loading dock?	A door opening inside an authorised depot or dock, at under 8 km/h, is logged as handling and becomes an alert only after 10 minutes outside limits. An open door never excuses a cooling hold.	none

Question	Answer	Slide
Auto-quarantine could hold good cargo.	A quarantine is a hold, not a loss. The administrator reviews it, and only a supervisor override with a written reason lifts it and issues a new custody token.	none
Admin approval slows the response.	The rules hold the cargo at once. Approval covers only disruptive steps such as diverting or taking the vehicle offline. How much time approval costs is not yet measured.	9
Will people get flooded with alerts?	A message goes out only when a problem is new, escalates, changes a package's status, or is still open after 10 minutes.	none
What if the node goes silent?	A watchdog alerts after 30 minutes without data, and after 10 minutes of stale GPS. The advice names the backup cold store.	none

4. AI and agentic claims

Question	Answer	Slide
Is this really agentic?	Each agent reads the situation, reasons over it and proposes actions from a fixed set of tools. Autonomy is deliberately limited, because a wrong disruptive action on medicine or food is costly. The human keeps the decision. State this as a design choice, not a gap.	4
What if the AI hallucinates?	0 invented numbers in 60 live calls, and a number check would drop any answer that had one.	8
Is 3 injection tests enough?	No, it is a demonstration. The real protection is structural: AI text can never set a status, release cargo or run an action.	8
Do you measure whether the explanations are good?	Not yet. The checks are automatic and narrow, covering invented numbers, menu violations, wrong claims and instructions obeyed. They do not score helpfulness.	8
Why Gemini free tier, and what if it fails?	It is enough for a prototype and limits calls to 15 a minute and 500 a day. If it fails or its answer is invalid, the rules' explanation is shown. A private endpoint is the next step.	none

5. Evidence and results

Question	Answer	Slide
Your results are simulated and written by you.	Agreed: ten scenarios, one author, and a deliberately fast baseline. They show what the rules do on known cases, not how often real incidents happen. Anyone can re-run them with judge_demo.bat, and a field trial is next.	none
Is the baseline fair?	It is a favourable one: an instant alarm with no delay, which is the fastest a threshold alarm can be.	none
What do the 312 tests prove?	The rules, the agents' safeguards, the web API and the document and standards outputs. They do not cover the physical board, which was checked by hand.	none
Is your EPCIS compliant or certified?	The events are valid ObjectEvents with hash IDs, checked against the official schema and hash vectors in tests. The company prefix is a demo and nothing is GS1-certified.	none
Is the sample incident report real?	It is an automatic draft from a simulated scenario, marked as not yet reviewed by a person. Its pictures are labelled sample frames.	none

6. Hardware and scale

Question	Answer	Slide
A DHT11 and a light sensor are crude for pharmaceuticals.	Yes, they are prototype parts: whole degrees, about 2 C accuracy, and light only approximates a door. Next: a calibrated probe, a reed-switch door sensor and a second thermometer.	10
What if there is no internet?	The node keeps one hour of readings and retries, and the rules run locally. Telegram, map tiles and Gemini need internet, and a real truck would need a gateway.	10
Does it scale to 1,000 trucks?	Not tested. Today it is one laptop and SQLite. The path is per-vehicle keys, a queue and a real database; one AI call per alert keeps cost bounded, but the free tier would need a paid plan.	none
Is it certified for regulated medicine?	No. Regulated transport needs calibrated, certified sensors. ColdTrust adds data trust and explanation on top of them.	none

7. About the team's work

Question	Answer	Slide
What was the hardest problem?	The evaluation caught that combining report wording and proposals in one call made 4 of 9 Incident replies run away, and that a stray control character could corrupt a Word file. You split the call and cleaned the text, with tests.	8
What would you do with another month?	A field trial on a real refrigerated truck, a second thermometer, GPS and a door switch, a private model endpoint, and measuring how much review time the reports save.	none

Words to avoid

- Do not say "unhackable", "guaranteed" or "100% accurate".
- Do not say your results are "proven" or from a "real deployment".
- Do not say "certified" or "fully compliant" about GS1 EPCIS.
- Do not say the AI "decides" anything; say it "explains and proposes".
- If you do not know, say "We have not measured that yet, and here is how we would", then give the next step.

Chapter 3: Video and preparation notes

These notes come from the Pitch Day Playbook and are still correct. Everything about the slide plan, the Canva checklist and the question list was replaced by Chapter 2.

The demo video: review and storyboard

Your video is 2:22 and tells the right story, but the first 33 seconds are talk, 22 seconds near the end have no narration, and it never states the measured result. Cut it to 2:00: the 5-minute slot includes the video, and a 2:30 video would leave only 2:30 for six slides. I read the full narration from YouTube's transcript; I could not see the picture, so the visual notes below follow your narration.

What each part does now

Time	What happens now	Verdict	Fix
0:00 to 0:33	Architecture talk before any action	Too long; judges form a view in the first 20 s	An 8 s hook over the hardware photo; move the architecture to slide 3
0:33 to 0:59	Ice on the sensor, alarm, voice, Telegram message from Frozen Warden	Strong: real hardware and a real phone	Keep, trim to 20 s, show the phone screen
0:59 to 1:16	Thermal excursion, depot stop with two handlers, report gate, all in 17 s	Three ideas in a rush	Keep the thermal explanation and the depot stop; turn the report gate into a caption
1:16 to 1:33	Spoofed thermometer: door open, 4 C flat, Not trusted	Your best moment and your main difference	Keep, zoom on the failed check, give it 25 s
1:33 to 1:50	Theft at 50 km/h, quarantine, tokens revoked, admin approves, driver cannot	Strong	Keep; show the driver view with no buttons for 3 s
1:50 to 2:12	Evidence photos and Word report; no narration for 22 s	Dead air	Cut to 12 s: report downloads, photos, hash
2:12 to 2:22	Caveats, then "This is ColdTrust"	No result, no call to action	Close on the result card and the GitHub line

Four things the video lacks: the problem in one sentence, the measured result (0 false alarms against 1, 3 of 3 attacks caught, 11 minutes earlier), a clear shot of the hardware, and burned-in captions. The automatic captions heard "ESB 8266", "HMAX" and "99913", which suggests the voice-over is hard to follow; add captions, or re-record it in short sentences.

New 2:00 storyboard

Time	Shot	Voice-over	On screen
0:00 to 0:08	Hardware photo, then the dashboard	"Cold chains fail when sensors drift or when the data is faked. ColdTrust checks its own data before it acts."	ColdTrust title
0:08 to 0:28	Ice on the DHT11, gauge falls, alarm, phone with the Telegram message	"Ice on the real sensor: freeze risk. The alarm sounds and Frozen Warden messages the team."	"Real hardware"

Time	Shot	Voice-over	On screen
0:28 to 0:48	Hold at 19 C with the Thermal agent text; depot stop with two handlers	"Each product has its own limit and budget, and the Thermal agent explains why. At an authorised dock, handling is not an alarm."	"Plain alarm: rings. ColdTrust: no alarm"
0:48 to 1:13	Spoof: door open, flat 4 C, Data trust card	"Now someone spoofs the thermometer. The door is open on a 31 C day but the temperature never moves. The Seal check marks the data Not trusted and shows which check failed."	Failed row highlighted
1:13 to 1:40	Theft at 50 km/h, quarantine, approvals, admin approves, driver view	"Theft in transit: the rules quarantine the cargo at once. The Incident agent proposes; only an admin can approve."	Admin and Driver labels
1:40 to 1:52	Evidence tab, Word report, hash	"A draft report with photo evidence and hashes is ready in seconds. A person reviews and signs."	"Draft, human sign-off"
1:52 to 2:00	Result card, then GitHub	"Against a plain alarm, in ten simulated scenarios: 0 false alarms against 1, 3 of 3 data attacks caught, first alert 11 minutes earlier."	"Simulated data. Try it: judge_demo.bat"

Clips and production

Most clips already exist in your current recording, so only three are new: the hardware hero shot (use `docs/evidence/hardware_overall.jpg`), a split screen of a plain alarm against ColdTrust at the dock stop, and the result card (the table from `docs/detection_results.md`).

- Record at 1080p with the browser at 125% zoom, and hide bookmarks and extra tabs.
- Burn captions into the video; export MP4 H.264 under 50 MB and embed the file in PowerPoint, never a link.
- Avoid anything that needs the internet during playback; the map tiles and Gemini both do.

Where each judging criterion is won

Criterion	Weight	Where in your final deck and video
Technical	25%	Slides 3 and 4, the spoof and theft clips, appendix slide 9, the 312 tests
Problem relevance	20%	Slide 2 (the research and CISA), the false-alarm result spoken on slide 6
Agentic AI	20%	Slide 4, the approval clip, appendix slide 8 (the AI evaluation)
Cybersecurity	20%	The spoof clip, slide 6, appendix slide 9 (Seal checks)
Presentation	15%	Timing, one idea per slide, a clean close on slide 7

Know, read and prepare

Numbers to know cold

Figure	Meaning	Where it comes from
0 against 1	False alarms, ColdTrust against a plain alarm	`docs/detection_results.md`

Figure	Meaning	Where it comes from
3 of 3 against 0 of 3	Data attacks recognised (replay, flat spoof, contradicting sensor)	`docs/detection_results.md`
5 of 5 against 0 of 5	Incidents with cargo held at the first alert	`docs/detection_results.md`
8:00 against 19:00	First alert on a slow cooler failure, 11 minutes earlier	`docs/detection_results.md`
60 of 60, 0, 0 of 3	Usable AI answers, invented numbers, injections obeyed	`docs/agent_eval.md`
1.6 s median, 7.1 s slowest	AI response time per call	`docs/agent_eval.md`
312	Automated tests passing	`docs/test_results.txt`
8, 4, 60	Seal checks, approval actions, readings (30 minutes) before a maintenance report	`backend/maintenance.py`, `backend/approvals.py`
1 h, 15 s	Approvals expire after an hour; theft report drafts about 15 s after the alert	`backend/approvals.py`, `backend/agent.py`
1 h	Node keeps this much data while Wi-Fi is down	`firmware/firmware.ino`

What to read before Saturday

- Your own limits: the "How to read this" part of `docs/detection_results.md` and the notes in `docs/agent_eval.md`; judges will probe exactly there.
- `REFERENCE.md` for roles, AI safeguards and privacy, and `PLAN.md` for why you designed it this way.
- CISA's December 2020 cold-chain guidance, and sections 4.4 and 4.6 of Abu Hassan et al. (2021). Know that the paper has no theft or security content; your security argument rests on CISA.
- A one-page GS1 EPCIS 2.0 introduction: what an ObjectEvent, bizStep and disposition are.
- A plain-words explanation of HMAC and replay attacks, so you can say it in two sentences.
- Google's current Gemini free-tier terms on data use.

Pitch-day checklist

☐ PPTX, PDF copy and MP4 on a USB stick and in cloud storage; open the PPTX on a second laptop and play the video.

☐ Charger, display adapter, clicker, phone hotspot.

☐ `judge_demo.bat spoof` already running in a minimised window as a live fallback; screenshots of the key screens in the appendix.

☐ Five timed run-throughs; aim to stop at 4:50.

☐ Print a one-page card of the numbers above.

☐ If more than one person answers questions, agree who takes technical, AI and security.